

Calibration Does Not Compose, Types Destroy Vagueness:

The Hidden-Markov and Fuzzy Primitives Missing from System-One Decision Models

Anh Khoa Doan Ngoc
khoa.doan@edge8.ai

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Abstract

System-one decision models—fast, non-generative networks that emit typed, calibrated probabilistic decisions for machine-to-machine pipelines, of which TypeSafe AI’s Jev, trained by Reinforcement Learning for Calibrated Decisions (RLCD), is the announced instance—are audited by hop-level calibration on stationary held-out data. We show that this class omits two structural primitives: recursive belief over a latent state (the hidden-Markov primitive) and graded predicate membership (the fuzzy primitive). We prove two negative results. Lemma 1: when outcomes are coupled through a latent Markov regime, marginal calibration is not preserved under regime shift, trajectory error counts are overdispersed relative to the binomial baseline that hop-level calibration implies, with a closed-form inflation factor, and every permutation-invariant audit statistic—expected calibration error included—has exactly zero power to detect this. Lemma 2: thresholding a typed score makes the composed pipeline action discontinuous; an ε -perturbation of evidence flips a full action on a set of measure $\Theta(\varepsilon)$, every downstream hop then receives an $O(1)$ input change so the conditional flip probability is $\Theta(1)$, and when several hops threshold the same coupled quantity the all-hops-flip event has measure $\Theta(\varepsilon)$ rather than $\Theta(\varepsilon^H)$; a fuzzy membership composed by a t-norm and defuzzified once by a Lipschitz map keeps the action Lipschitz in the evidence. We state the Principle of Deferred Crispification and specify a reference architecture, Belief-State Fuzzy System-One (BSF-S1), whose output object carries a per-regime membership matrix, a calibrated distribution over the degree, and a scaled-likelihood vector over regimes, and collapses exactly once at the actuator at $O(K^2 + KM)$ cost per step. Five CPU-scale experiments confirm each prediction, and two operationally defined audit metrics (trajectory calibration error and action-mass sensitivity) can be run against a closed model from the outside. The paper is architectural: Jev is closed, and nothing here is an empirical claim about its internals.

1 Introduction

1.1 The rise of system-one decision models

For most of the last five years the dominant interface to machine intelligence has been text: a prompt goes in, a string comes out, and software that wants a decision must parse, validate and threshold that string. On 15 September 2026 TypeSafe AI released Jev, which its authors describe as “a frontier-intelligence function call: unstructured state in, typed probabilistic decisions out” (Almeida, 2026). Jev generates no text. Its output schema is fixed in advance through three primitives—*Choice* (an enumeration with a probability attached to each option), *Score* (a bounded scalar) and *Noul*—and every answer is accompanied by a confidence score; the documentation’s own example is a ticket-routing *Choice* over **billing**, **technical** and **account**, returned with a probability per option (TypeSafe AI, 2026). All outputs of a query are produced in a single parallel pass rather than autoregressively, which is where the speed comes from:

end-to-end latency of 70–500 ms, “40×–200× faster” than frontier language models, input priced at \$0.042 per million tokens and output “too cheap to meter” (Almeida, 2026). The launch drew immediate trade-press coverage (Claburn, 2026); every technical statement in this paper is taken from TypeSafe’s own launch post and documentation. The training method is named Reinforcement Learning for Calibrated Decisions (RLCD) and is said to optimise for “answers with epistemically honest probabilities” (Almeida, 2026).

We take this launch as the concrete instance of a class we call *system-one decision models*: fast, memoryless, typed, calibrated decision heads intended to be consumed by software rather than by people. The class is new enough that its audit vocabulary has been borrowed wholesale from the classification literature. This paper asks whether that vocabulary is adequate.

1.2 The audit practice of the field

The audit practice is hop-level. A model is scored on a held-out sample of (score, outcome) pairs drawn from the same distribution as training; expected calibration error (ECE) or a proper scoring rule summarises the gap between stated and realised frequencies; accuracy benchmarks summarise per-decision correctness. TypeSafe’s documentation states the scope of this measurement with admirable precision: “Calibration is measured across groups of predictions; it does not guarantee that an individual answer is correct” (TypeSafe AI, 2026). This paper takes that sentence seriously and asks a further question: which *groups* is the measurement silent about?

1.3 Problem statement

We identify two omissions.

- (i) **The posterior is memoryless.** The model emits $P(d \mid o)$, calibrated against a frozen marginal prior over the world’s latent state. There is no recursion that carries belief about that state from one decision to the next. When the world’s regime changes, the score remains “calibrated” with respect to a prior that no longer exists.
- (ii) **The output type is crisp.** A typed enumeration commits at design time to sharp boundaries between predicates that, for a positive fraction of inputs, are intrinsically borderline. The model is forced to express ontological vagueness (“this ticket is *somewhat* urgent”) as epistemic uncertainty (“I am 50% sure it is urgent”), and the harness reads the latter as ignorance.

1.4 Thesis

Both omissions are *structural*: they concern what object the model emits and how the harness composes it, not the capacity of the network that produces the object. They survive any scaling of the network, any improvement in the RL objective, and any amount of additional training data, because they are properties of the type signature of the output. They are also *cheap to repair*: the missing machinery is a K -state forward filter and an M -dimensional membership head, costing $O(K^2 + KM)$ operations per step, negligible against a single encoder pass and requiring no generative decode.

1.5 Contributions

- C1** A diagnosis of the temporal hole and Lemma 1 (calibration does not compose): prior-ratio mismatch under regime shift; overdispersion of trajectory error counts with a closed-form inflation factor; and a proof that every permutation-invariant audit statistic is blind to the effect.

- C2** A diagnosis of the vagueness hole and Lemma 2 (cliff cascade): discontinuity of the thresholded pipeline, $\Theta(\varepsilon)$ flip mass with $O(1)$ jumps, $\Theta(1)$ conditional cascade probability, $\Theta(\varepsilon)$ all-hops-flip under coincident thresholds, and Lipschitz continuity of the fuzzy alternative.
- C3** The Principle of Deferred Crispification.
- C4** BSF-S1, a reference architecture with a specified output object, a composition algebra for each component, an exact forward filter over the scaled-likelihood component, and a single actuator collapse.
- C5** An experimental program and two audit metrics: trajectory calibration error (TCE) and action-mass sensitivity (AMS).

2 Background and Related Work

2.1 System-one decision models and RLCD

What is public about Jev (Almeida, 2026; TypeSafe AI, 2026) is the following. The model “evaluates a state and returns typed answers and probabilities”; its probabilities “are optimized against outcomes to reflect uncertainty”; it “does not write replies, produce code, or generate explanations of its reasoning”; and all outputs of a query are generated in parallel. Two further statements in the launch material are load-bearing for this paper. First, the recommended integration pattern is to “ask independent questions together . . . combine the answers with deterministic checks in code, then route the case” (TypeSafe AI, 2026): the harness thresholds the returned probabilities. Second, the authors observe that “the most reliable real-world workflows tend to have many independent, decomposed questions, with fine-grained behavior that’s dependent on probabilities instead of discrete decisions. The end result is discrete branching” (Almeida, 2026). This is an accurate description of the pipeline whose composition properties Sections 3 and 4 analyse. The launch material also claims consistency—the model “returns similar answers for similar inputs”—which is a Lipschitz-type property of the *score*; Lemma 2 concerns what happens to that property once the score is thresholded.

Jev’s weights, architecture and training data are closed. Every statement in this paper is about the *class* of memoryless typed calibrated heads, of which Jev is the announced instance; nothing here is an empirical claim about Jev’s internals.

2.2 Hidden Markov models and Bayesian filtering

Let $s_t \in \{1, \dots, K\}$ be a latent regime following a Markov chain with transition matrix $A \in \mathbb{R}^{K \times K}$, $A_{ij} = P(s_{t+1} = j \mid s_t = i)$, and stationary distribution π with $\pi^\top A = \pi^\top$. Evidence o_t is emitted with $P(o_t \mid s_t)$. The forward filter (Rabiner, 1989)

$$b_t(s) \propto P(o_t \mid s) \sum_{s'} A_{s's} b_{t-1}(s') \quad (1)$$

is exact inference given (A, B) . Regime-switching econometrics (Hamilton, 1989) is the same recursion with Gaussian emissions; partially observable Markov decision processes (Kaelbling et al., 1998) add actions. The hybrid HMM/neural-network literature (Bourlard and Morgan, 1994) established that a discriminatively trained network emitting $P(s \mid o)$ can serve as an emission model once its outputs are “normalized by class prior probabilities to provide scaled likelihoods” $P(o \mid s)/P(o) = P(s \mid o)/\pi(s)$. This is the bridge used in Section 3.5.

2.3 Fuzzy sets and fuzzy logic

A type-1 fuzzy set on X is a membership function $\mu : X \rightarrow [0, 1]$ (Zadeh, 1965). Conjunction and disjunction are *truth-functional*: the degree of “ A and B ” is $T(\mu_A, \mu_B)$ for a t-norm T (minimum, product, Łukasiewicz), and the degree of “ A or B ” is a t-conorm S (Klement et al., 2000). Minimum is the unique idempotent t-norm. Mamdani control (Mamdani and Assilian, 1975) evaluates a rule base, aggregates the consequents into an output membership, and defuzzifies *once*, at the plant input, typically by centroid. A type-2 fuzzy set (Zadeh, 1975; Mendel, 2007) attaches to each x a distribution over the degree, separating “how much x is M ” from “how sure we are about how much.”

2.4 Calibration and proper scoring

A score f is marginally calibrated under P if $\mathbb{E}_P[Y \mid f(X) = p] = p$ for all p . ECE is the binned L^1 gap between this conditional mean and p (Guo et al., 2017). A scoring rule is strictly proper if the true distribution uniquely minimises the expected score (Gneiting and Raftery, 2007). Calibration of a marginal says nothing about the joint distribution of several decisions; that distinction is the entire content of Section 3.

2.5 Scope and non-goals

We make no claims about generative models, network architecture, scale, context length or caching; no claims about cognition (the “system-one” label is Kahneman’s framing (Kahneman, 2011) only). We study the decision *output object* and its *composition* across hops.

3 Analysis I: The Temporal Hole

3.1 The memoryless calibrated posterior

Let the world have latent regime s_t as in Section 2, evidence o_t with $P(o_t \mid s_t)$, and decision target $d_t \in D$ with $P(d_t \mid o_t, s_t)$. The system-one head is $f(o) \approx P_{\text{tr}}(d \mid o)$, where the training distribution mixes regimes with their stationary weights:

$$P_{\text{tr}}(d \mid o) = \sum_s P(d \mid o, s) P_{\text{tr}}(s \mid o), \quad P_{\text{tr}}(s \mid o) \propto P(o \mid s) \pi(s). \quad (2)$$

An RLCD-style objective can certify that f is calibrated under P_{tr} . It cannot certify anything about $P(d \mid o)$ under a distribution in which the regime weights differ, because the objective never observes s and never observes more than one time step at once.

3.2 Regime shift as prior death

Suppose that at serving time the regime marginal is $\pi' \neq \pi$: a sojourn in a rare regime, or a permanent drift. The true conditional becomes

$$P'(d \mid o) = \sum_s P(d \mid o, s) P'(s \mid o), \quad P'(s \mid o) \propto P(o \mid s) \pi'(s). \quad (3)$$

The head still outputs (2). Its score is “correct” with respect to a prior that has died. A held-out set drawn from P_{tr} has calibration error identically zero for a perfectly trained f , so the audit is structurally blind to (3). This is not a failure of training; it is the absence of a variable.

3.3 Lemma 1

Lemma 1 (Calibration does not compose). *Let $d \in \{0, 1\}$ (the K -ary case is identical componentwise). Assume the HMM conditional independences $o_t \perp o_{t'} \mid s_t, s_{t'}$ and $d_t \perp (o_{t'}, d_{t'}) \mid (s_t, o_t)$ for $t \neq t'$. Let f be perfectly calibrated under P_{tr} . Let $E_t = \mathbb{1}\{\text{decision at } t \text{ is wrong}\}$, $e_s = P(E_t = 1 \mid s_t = s)$ the regime-conditional error rate of the fixed decision rule, and $\bar{e} = \sum_s \pi(s) e_s$.*

(i) (Prior-ratio mismatch.) *Under serving marginal π' , the pointwise miscalibration of f is*

$$\Delta(o) := P'(d = 1 \mid o) - f(o) = \sum_s P(d = 1 \mid o, s) [P'(s \mid o) - P_{\text{tr}}(s \mid o)], \quad (4)$$

nonzero whenever the regime-conditional answers differ across s and $\pi' \neq \pi$. The ECE under P' is bounded below by $|\mathbb{E}_{P'} \Delta(o)|$ while the ECE under P_{tr} is zero.

(ii) (Overdispersion.) *For a two-regime chain $A = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ with $\lambda := 1-a-b \in (-1, 1)$,*

$$\text{Cov}(E_t, E_{t+k}) = \lambda^k \text{Var}_{\pi}(e), \quad \text{Var}_{\pi}(e) := \sum_s \pi(s) e_s^2 - \bar{e}^2. \quad (5)$$

Hence the window error count $N_T = \sum_{t=1}^T E_t$ satisfies

$$\text{Var}(N_T) = T \bar{e}(1 - \bar{e}) + 2 \text{Var}_{\pi}(e) \sum_{k=1}^{T-1} (T-k) \lambda^k, \quad (6)$$

and the asymptotic variance-inflation factor over the binomial baseline implied by hop-level calibration is

$$\varphi := \lim_{T \rightarrow \infty} \frac{\text{Var}(N_T)}{T \bar{e}(1 - \bar{e})} = 1 + \frac{2\lambda}{1 - \lambda} \cdot \frac{\text{Var}_{\pi}(e)}{\bar{e}(1 - \bar{e})}. \quad (7)$$

$\varphi > 1$ iff $\lambda > 0$ and $\text{Var}_{\pi}(e) > 0$. Error runs have the lengths of regime sojourns, geometric with means $1/a$ and $1/b$. For a K -regime chain with spectral decomposition $A^k = \mathbf{1}\pi^\top + \sum_{i \geq 2} \lambda_i^k P_i$, (5) becomes $\text{Cov}(E_t, E_{t+k}) = \sum_{i \geq 2} \lambda_i^k e^\top \text{diag}(\pi) P_i e$.

(iii) (All-correct probability.) *$P(N_T = 0) = \mathbb{E}[\prod_t (1 - e_{s_t})] \geq (1 - \bar{e})^T$ whenever $\lambda \geq 0$, strictly iff $\lambda > 0$ and $\text{Var}_{\pi}(e) > 0$. The trajectory distribution is heavier in both tails than the independent baseline.*

(iv) (Permutation-invariance blindness.) *ECE, any proper score averaged over hops, and any statistic that is a function of the empirical distribution of $(f(o_t), y_t)$ pairs is invariant under permutation of t . A clustered sequence and its random shuffle have identical such statistics while (6) differs. Hop-level audits therefore have exactly zero power against (ii)–(iii).*

Proof sketch. (i) is (2) subtracted from (3). (ii): condition on (s_t, s_{t+k}) ; by the independences $\mathbb{E}[E_t E_{t+k}] = e^\top \text{diag}(\pi) A^k e$; substitute $A^k = \mathbf{1}\pi^\top + \lambda^k (I - \mathbf{1}\pi^\top)$, valid for every 2×2 stochastic matrix; the first term gives $\bar{e}^2$ and the second $\lambda^k \text{Var}_{\pi}(e)$. Summing covariances gives (6); the geometric series gives (7). (iii): $x \mapsto x^T$ is convex on $[0, 1]$; at $\lambda = 1$ Jensen gives the inequality directly, and for $0 < \lambda < 1$ the two-state chain is the mixture $A = (1 - \lambda)\mathbf{1}\pi^\top + \lambda I$, whose paths consist of independent frozen blocks. (iv): ECE is a function of the multiset of pairs, and multisets are permutation-invariant. Full proofs are in Appendix B. $\square$

Numerical check. With $a = 0.05$, $b = 0.10$, $e = (0.05, 0.40)$ and $T = 50$, simulation over 2×10^5 trajectories gives $\text{Var}(N_T) = 20.30$ against 20.31 from (6) and 6.94 for the binomial baseline; $P(N_T = 0)$ is 0.0071 against 0.0001 for the independent baseline. The script is in the repository.

3.4 Consequence

Risk in a pipeline is a trajectory-level object. The quantity a downstream consumer cares about is the distribution of N_T —how many wrong decisions arrive in this window, and whether they arrive together—not $\bar{e}$. Hop-level calibration certifies $\bar{e}$ and is silent on φ . Two models with identical ECE can differ in φ by an arbitrary factor, and a consumer that budgets for binomial error is under-provisioned by exactly that factor.

3.5 Repair primitive

Treat the network as an emission model and run the exact filter.

1. Train a regime head $g(o) \approx P_{\text{tr}}(s \mid o)$ (K -way softmax) alongside per-regime decision heads $h_s(o) \approx P(d \mid o, s)$.
2. Convert to a scaled likelihood: $\Delta b_t(s) := g_s(o_t)/\pi(s)$, normalised to sum to one. The omitted normaliser is a function of o_t alone and cancels in (8).

3. Filter:

$$\tilde{b}_t(s) = \sum_{s'} A_{s's} b_{t-1}(s'), \quad b_t(s) = \frac{\tilde{b}_t(s) \Delta b_t(s)}{\sum_{s''} \tilde{b}_t(s'') \Delta b_t(s'')}. \quad (8)$$

4. Decide: $P(d_t \mid o_{1:t}) = \sum_s h_s(o_t) b_t(s)$.

Given a correct (A, g, h) the decision posterior is the true conditional, hence calibrated on any window under any π' , and its implied joint over N_T is the true joint: trajectory calibration is a consequence of exact inference, not an additional objective. The reason the carried object is a scaled likelihood and not a posterior is that two posteriors $P(s \mid o_1)$ and $P(s \mid o_2)$ each already contain π ; multiplying them counts π twice. Dividing each by $\pi(s)$ leaves objects proportional to $P(o_i \mid s)$, which do multiply under conditional independence.

4 Analysis II: The Vagueness Hole

4.1 Two distinct “not-sures”

Epistemic uncertainty: the world is in exactly one crisp state, we do not know which, and a probability over states sums to one. Ontological vagueness: the predicate itself has no sharp boundary; membership is a degree; degrees over different predicates need not sum to anything. A single number in $[0, 1]$ attached to an enumeration value conflates the two, and every consumer of that number must guess which one it is receiving.

4.2 Typed enumerations as frozen ontology commitments

A schema `priority` $\in \{\text{low}, \text{medium}, \text{high}\}$ commits at design time to sharp boundaries the data do not have. An input that is genuinely borderline between medium and high is best described by memberships $(0, 0.5, 0.5)$. A calibrated probabilistic head trained on inconsistent labels for such inputs learns $p \approx (0, 0.5, 0.5)$ as a *probability*, which the harness reads as “the model does not know” and routes to escalation, retry, or a more expensive model. The information “this is a borderline case and will remain one no matter how many times you ask” is destroyed, and the cost of that destruction is paid on every borderline input forever. The Score primitive in Jev’s schema (TypeSafe AI, 2026) can express a degree as a bounded scalar; but a single scalar cannot express both the degree and the uncertainty about the degree, which is the type-2 distinction of Section 4.5.

4.3 Lemma 2

Lemma 2 (Cliff cascade). *Let evidence $x \in X \subset \mathbb{R}^n$ have distribution P_X . There are H hops; hop h has an L -Lipschitz score $f_h : X \rightarrow [0, 1]$ and threshold $\tau_h \in (0, 1)$. Assume the pushforward of P_X under f_h has a density p_h positive and finite at τ_h , and $\|\nabla f_h\| \geq c > 0$ on $f_h^{-1}(\tau_h)$.*

Crisp pipeline: $a_h(x) = \mathbb{1}\{f_h(x) \geq \tau_h\}$ and outcome $G(x) = \bigwedge_h a_h(x)$ (any non-constant Boolean combination gives the same conclusions). Sequential crisp pipeline: hop $h+1$'s evidence includes a_h , $x_{h+1} = (x, a_h)$. Fuzzy pipeline: $\mu_h = f_h$ read as a membership, composed by a t -norm T that is 1-Lipschitz in ℓ^1 on $[0, 1]^H$ (minimum and product both are), and defuzzified once by an L_D -Lipschitz map D ; action $u(x) = D(T(\mu_1(x), \dots, \mu_H(x)))$.

- (i) (Discontinuity vs. Lipschitz.) G is discontinuous on $\bigcup_h f_h^{-1}(\tau_h)$, whose ε -neighbourhood has positive P_X -measure for every $\varepsilon > 0$. The fuzzy action satisfies, for all x, x' ,

$$|u(x) - u(x')| \leq L_D \sum_h |\mu_h(x) - \mu_h(x')| \leq L_D H L \|x - x'\|. \quad (9)$$

- (ii) ($\Theta(\varepsilon)$ flip mass with $O(1)$ jumps.) Let $F_\varepsilon = \{x : \exists x', \|x' - x\| \leq \varepsilon, G(x') \neq G(x)\}$. Then $P_X(F_\varepsilon) = \Theta(\varepsilon)$, with per-hop coefficient between $c p_h(\tau_h)$ and $2L p_h(\tau_h)$. On F_ε the worst-case action change is 1. For the fuzzy pipeline the worst-case action change is at most $L_D H L \varepsilon$ at every x ; there is no set on which it is $O(1)$.

- (iii) (Cascade.) In the sequential crisp pipeline, conditional on a_h flipping, hop $h+1$'s evidence changes by $\delta_h = \|(x, 1) - (x, 0)\|$, independent of ε , so

$$P(a_{h+1} \text{ flips} \mid a_h \text{ flips}) \geq P_X(|f_{h+1} - \tau_{h+1}| \leq c \delta_h) =: q_{h+1} = \Theta(1). \quad (10)$$

The expected number of downstream flips triggered by one ε -perturbation, $\sum_h \prod_{j \leq h} q_j$, does not vanish as $\varepsilon \rightarrow 0$. In the sequential fuzzy pipeline hop $h+1$ receives μ_h , whose change is at most $L\varepsilon$, so the total action change is at most $L_D(\sum_h L^h)\varepsilon$: geometric in L , still $O(\varepsilon)$.

- (iv) (Common-mode coupling.) Suppose the hops' scores are monotone functions of a shared latent scalar z , $f_h(x) = \phi_h(z, \xi_h)$, and the perturbation acts on z . If the hops' crossing points coincide—there is a z^* with $\phi_h(z^*, \xi_h) = \tau_h$ for every h , as when several hops threshold the same score (a model threshold followed by a harness threshold) or evaluate the same regime-dependent predicate—then $P(\text{all } H \text{ hops flip}) = \Theta(\varepsilon)$. If instead the crossing points are spread with a continuous joint density, $P(\text{all flip}) = \Theta(\varepsilon^H)$ even under a common-mode perturbation, the same order as under independent per-hop perturbations. Coupling therefore bites through coincident thresholds, which is precisely the double-collapse pattern of Section 5.

Proof sketch. (i) A step function composed with a continuous map is discontinuous on the preimage of the threshold; (9) chains Lipschitz bounds using $|\prod_h a_h - \prod_h b_h| \leq \sum_h |a_h - b_h|$ on $[0, 1]^H$ (telescoping) and $|\min a - \min b| \leq \max_h |a_h - b_h|$. (ii) F_ε contains the band $\{\tau_h - c\varepsilon \leq f_h < \tau_h\}$, reachable by moving ε along ∇f_h , and is contained in $\{|f_h - \tau_h| \leq L\varepsilon \text{ for some } h\}$ by Lipschitzness; both have mass $\Theta(\varepsilon)$ by the density assumption. (iii) is (10). (iv) With coincident crossing points the set of z within ε of z^* has measure $\Theta(\varepsilon)$ and monotonicity forces simultaneous crossing; with spread crossing points all H of them must lie within a 2ε window that z then enters, an event of order $\varepsilon^{H-1} \cdot \varepsilon$. Appendix B. $\square$

Remark 1 (On the defuzzifier). Inequality (9) requires D Lipschitz. The centroid $D(\mu_{\text{out}}) = \int y \mu_{\text{out}}(y) dy / \int \mu_{\text{out}}(y) dy$ is Lipschitz in μ_{out} (sup norm) on the set $\{\int \mu_{\text{out}} \geq m_0 > 0\}$; the architecture must enforce the floor m_0 (a regularised centroid). Mean-of-maxima and height defuzzifiers are *not* continuous and would reintroduce the cliff. The principle of Section 5 therefore names the defuzzifier.

Remark 2 (What the lemma does not claim). For a smooth score distribution, the *expected* L^1 sensitivity of the first hop is $O(\varepsilon)$ in both pipelines. The separation is in the per-point worst case (a jump of 1 versus a bound of $L\varepsilon$), in the cascade (a $\Theta(1)$ conditional flip probability versus $O(\varepsilon)$ propagation), and in coincident thresholds ($\Theta(\varepsilon)$ versus $\Theta(\varepsilon^H)$). Readers who have seen “fuzzy is smoother” asserted without qualification should note that (iii) and (iv) are where the content lies.

4.4 The wrong algebra

Probability is not truth-functional: $P(A \wedge B)$ is not determined by $P(A)$ and $P(B)$ without an independence assumption. A t-norm is truth-functional by construction: $T(\mu_A, \mu_B)$ is the degree of “ A and B .” When a harness multiplies typed probabilities across hops it silently asserts independence of predicates that are typically about the same object; when it takes a maximum it asserts nothing coherent. The product t-norm coincides numerically with independent multiplication but is licensed by a different semantics. Minimum is the unique idempotent t-norm (Klement et al., 2000), and idempotence is exactly the property that prevents a predicate from being double-counted when it enters two rules: product gives μ^2 , which is the origin of the “confident by repetition” failure in which asking the same question twice manufactures certainty.

4.5 Repair primitive

Emit a membership vector $\mu(x) \in [0, 1]^M$ over the M linguistic predicates the schema cares about, with no sum constraint, and a type-2 layer $Q[\mu]$ —a distribution over $[0, 1]^M$, in practice per-predicate Beta parameters—carrying calibrated uncertainty about the degree. Calibration of Q is quantile calibration, $P(\mu_{\text{true},m} \leq Q_m^{-1}(\alpha)) = \alpha$, audited by the continuous ranked probability score or the pinball loss against elicited degrees. The two axes are never merged.

5 The Principle of Deferred Crispification

5.1 Three structures, three algebras

Structure	Object	Composition algebra	Audit
Vagueness	$\mu \in [0, 1]^M$	t-norm / t-conorm, rule base	agreement with elicited degrees
Temporal coupling	$\Delta b \in \Delta^K$ (scaled likelihood)	pointwise product with predicted belief	trajectory calibration (TCE)
Uncertainty about degree	$Q[\mu]$	mixture / posterior update	proper scores (CRPS, pinball)

Table 1: Each structure has its own object, its own composition rule and its own audit. Collapsing any of them early substitutes the wrong algebra for the right one.

5.2 Statement

Principle 1 (Deferred Crispification). No structure may be collapsed before the actuator. Collapse—defuzzify, threshold, sample—happens exactly once, with full decision context: the belief over regimes, the cost matrix, and every membership in hand.

Each early collapse is a specific information loss. Thresholding μ discards the degree (Lemma 2). Marginalising b_t discards the regime (Lemma 1). Collapsing Q to its mean discards whether the degree is known. None of these losses can be recovered downstream, because the downstream consumer receives only the collapsed object.

5.3 Output object

$$O(x_t) = (\mu_t \in [0, 1]^{K \times M}, \quad Q_t, \quad \Delta b_t \in \Delta^K). \quad (11)$$

μ_t is emitted *per regime*: row s is the membership vector the model would report if the regime were s . Q_t likewise. This is the fully deferred form; mixing over regimes is itself a collapse and is left to the actuator. When $K \times M$ is too large, the harness may pass the belief-weighted mixture $\bar{\mu}_t = \sum_s b_t(s) \mu_t^{(s)}$ —a convex combination, hence still a membership vector—at the cost of an early collapse over s , which must be declared.

5.4 Interface contract for harnesses

Downstream hops consume O uncollapsed. Rule bases compose μ with t-norms and t-conorms and never threshold. Filters compose Δb with A by (8) and never receive a posterior from another hop, because a posterior would double-count π . Scorers audit Q with proper scores and audit the filter with TCE. Only the actuator collapses.

5.5 Precedent audit

Three independent engineering traditions arrived at deferred crispification in their own dialects, and their agreement is the strongest evidence we can offer for the principle short of the experiments of Section 7.

Fuzzy control. Since Mamdani’s steam-engine controller (Mamdani and Assilian, 1975), the industrial pattern has been: fuzzify the sensor reading, evaluate every rule to a firing degree, aggregate the consequents into one output membership, and defuzzify once at the plant input. Fifty years of deployment in process control, consumer appliances and automotive systems have not produced a successful variant that thresholds inside the rule base. The reason is Lemma 2: a controller that thresholds early chatters at the boundary.

Probabilistic forecasting. Meteorology and finance learned in the 1990s to deliver full predictive distributions to the decision-maker and to audit them with proper scoring rules (Gneiting and Raftery, 2007), precisely because a point forecast collapsed at the model discards the information a cost-sensitive consumer needs. The ensemble is carried to the head; the decision is made there.

Regime-switching econometrics and POMDP planning. Hamilton’s filter (Hamilton, 1989) carries the filtered regime probability through time and never commits to a point regime; POMDP planners (Kaelbling et al., 1998) act on the belief state, not on its argmax. In both, the belief is the carried object and the collapse is the action.

Contemporary agent stacks violate the principle twice. The model thresholds (or the harness argmaxes the returned distribution), and the harness thresholds again on the resulting booleans. Jev’s integration guidance—combine typed probabilities “with deterministic checks in code, then route” (TypeSafe AI, 2026)—is this pattern stated as best practice. Lemma 2(iii) is the analysis of what the second threshold does to the first.

6 Architecture: Belief-State Fuzzy System-One

6.1 Emission block

One shared encoder $\phi(o_t)$ feeds: a regime head $g(o_t) \in \Delta^K$; per-regime decision heads $h_s(o_t) \in \Delta^{|D|}$; per-regime membership heads $\mu^{(s)}(o_t) \in [0, 1]^M$ (sigmoid, no softmax); and per-regime type-2 heads emitting Beta parameters $(\alpha_m^{(s)}, \beta_m^{(s)})$ so that $Q_m^{(s)} = \text{Beta}(\alpha, \beta)$ has mean $\mu_m^{(s)}$.

6.2 Transition block

$A \in \mathbb{R}^{K \times K}$ row-stochastic, parameterised by row-wise softmax of a free matrix. We keep the probabilistic reading so that (8) is exact Bayes.

6.3 Update block and an open problem

Predict and update by (8) with $\Delta b_t(s) = g_s(o_t)/\pi(s)$, π the stationary distribution of the current A . The belief b_t renormalises because it is a probability over a partition; the memberships $\mu_t^{(s)}$ do not, because predicates are not a partition. The belief-weighted mixture $\bar{\mu}_t$ is mathematically well-defined, but whether “membership in M under uncertainty about which regime’s M applies” should be the mixture, the minimum over plausible regimes, or Zadeh’s probability of a fuzzy event is a semantic question we state and do not resolve. BSF-S1 sidesteps it by passing the per-regime matrix and letting the actuator choose.

6.4 Propagation and actuator

Pass $O(x_t)$ and, for convenience, b_t itself flagged as *not composable*. The actuator performs the single collapse with context: belief-weighted mixture over regimes, t-norm aggregation across the rule base, regularised centroid defuzzification, cost-sensitive decision, optional sampling.

6.5 Algorithms

Algorithm 1 BSF-S1 step (per observation; $O(K^2 + KM)$ beyond the encoder)

Require: $o_t, b_{t-1} \in \Delta^K$, A , π , encoder ϕ , heads (g, h, μ, Q)

- 1: $z \leftarrow \phi(o_t)$
- 2: $\Delta b \leftarrow \text{normalise}(g(z)/\pi)$ ▷ scaled likelihood
- 3: $\tilde{b} \leftarrow A^\top b_{t-1}$ ▷ predict, $O(K^2)$
- 4: $b_t \leftarrow \text{normalise}(\tilde{b} \odot \Delta b)$ ▷ update, $O(K)$
- 5: $S_t \leftarrow -\log \sum_s \tilde{b}[s] \Delta b[s]$ ▷ belief surprise
- 6: **for** $s = 1, \dots, K$ **do** ▷ $O(KM)$
- 7: $\mu_t[s, :] \leftarrow \mu^{(s)}(z); \quad Q_t[s, :] \leftarrow (\alpha^{(s)}(z), \beta^{(s)}(z)); \quad P_t[s, :] \leftarrow h_s(z)$
- 8: **end for**

Ensure: $O_t = (\mu_t, Q_t, \Delta b), b_t, P_t, S_t$

Algorithm 2 Actuator: the single collapse

Require: O_t, b_t, P_t ; rule base $\mathcal{R} = \{(\text{ante}_r, C_r)\}$; cost matrix $\text{Cst} \in \mathbb{R}^{|D| \times |D|}$; t-norm T ; floor

- 1: $\bar{\mu} \leftarrow \sum_s b_t[s] \mu_t[s, :]; \quad \bar{P} \leftarrow \sum_s b_t[s] P_t[s, :]$ ▷ the only marginalisation over s
- 2: **for all** rules r **do**
- 3: $w_r \leftarrow T(\bar{\mu}[m] : m \in \text{ante}_r)$ ▷ firing degree; no threshold
- 4: **end for**
- 5: $\mu_{\text{out}}(y) \leftarrow \max_r \min(w_r, C_r(y))$ ▷ Mamdani aggregation
- 6: $\text{area} \leftarrow \int \mu_{\text{out}}; \quad \text{if } \text{area} < m_0: \mu_{\text{out}} \leftarrow \mu_{\text{out}} + (m_0 - \text{area})/|Y|$ ▷ regularised centroid
- 7: $u \leftarrow \int y \mu_{\text{out}}(y) dy / \int \mu_{\text{out}}(y) dy$ ▷ continuous actuator signal
- 8: $d^* \leftarrow \arg \min_d \sum_{d'} \text{Cst}[d, d'] \bar{P}[d']$ ▷ cost-sensitive decision
- 9: $\bar{Q} \leftarrow \sum_s b_t[s] Q_t[s, :]$ ▷ kept as a mixture for the consumer

Ensure: $(u, d^*, \bar{Q})$

Algorithm 3 Training

- 1: **Stage 0.** Pretrain ϕ and g on regime labels if available; otherwise choose K by BIC over an HMM fitted on encoder features.
 - 2: **Stage 1 (EM, heads fixed).** Baum–Welch for A with Δb_t as emission terms (scaled-likelihood HMM).
 - 3: **Stage 2 (joint).** The recursion (8) is differentiable. Minimise

$$\mathcal{L} = -\sum_t \log \sum_s h_s(o_t)[d_t] b_t(s) + \lambda_\mu \sum_{t,m} \text{CRPS}(Q_{t,m}, \mu_{t,m}^{\text{elicited}}) + \lambda_A \text{KL}(\text{rows of } A \parallel \text{prior}).$$

$P(o_{1:T})$ is unavailable with scaled likelihoods; the decision NLL through the filter is the computable surrogate.
 - 4: **Stage 3.** Temperature-scale g and each h_s per regime on held-out windows; verify TCE ≈ 0 before deployment.
-

Serving cost. Beyond the encoder, Algorithm 1 costs $O(K^2 + KM)$ multiply-adds and Algorithm 2 costs $O(|\mathcal{R}|M + |Y|)$. With $K \leq 16$ and $M \leq 64$ this is under 10^4 operations, orders of magnitude below one encoder pass, with no generative decode. The architecture is system-one compatible by construction.

6.6 Failure-mode inventory

- *Filter misspecification.* Wrong A or g gives exact inference over a wrong model. The belief surprise S_t of Algorithm 1 has a known expectation under the model; a CUSUM on $S_t - \mathbb{E}[S_t]$ raises a drift alarm.
- *K identifiability.* Regimes are latent; K is a modelling choice. Over-specified K degrades gracefully; under-specified K reintroduces Lemma 1 within a merged regime.
- *Membership ground truth.* Elicitation-dependent (Section 8); Q is trained to absorb elicitation disagreement as degree-uncertainty rather than pushing μ toward 0.5.
- *Stale A under drift.* Detected by the surprise monitor; repaired by incremental Stage 1.

7 Experiments

All experiments run on a laptop CPU in about a minute, use only NumPy and SciPy, and are reproduced by one command in the repository. Models are deliberately tiny—logistic heads on quadratic features and a two-regime filter—because every claim concerns the output object and its composition, not capacity.

7.1 Operational definitions of the two audit metrics

Definition 1 (Trajectory calibration error, TCE_w). Fix a window length w . Every decision model implies a joint predictive distribution for the window error count $N_w = \sum_{t \in \text{window}} E_t$: a memoryless head implies the Poisson-binomial distribution of independent Bernoulli($1 - \max(p_t, 1 - p_t)$) errors, which is the only joint it can express; BSF-S1 implies the distribution obtained by forward-filtering, backward-sampling regime paths (Appendix C) and drawing errors from the per-regime heads. Let F_i be the implied CDF for window i and n_i the realised count. The randomised probability integral transform is $u_i \sim \text{Unif}[F_i(n_i - 1), F_i(n_i)]$. TCE_w reports (a) the Kolmogorov–Smirnov p -value of $\{u_i\}$ against $\text{Unif}[0, 1]$ and (b) the dispersion ratio $\hat{\varphi} = \text{Var}_{\text{realised}}(N_w) / \text{Var}_{\text{implied}}(N_w)$, where the implied variance is the model’s total variance (within-window variance plus variance of window means). A model *passes* if KS does not reject at $\alpha = 0.01$ and $\hat{\varphi} \in [0.8, 1.25]$.

Definition 2 (Action-mass sensitivity, $\text{AMS}(\varepsilon)$). Fix a perturbation model: *adversarial*, $\delta = \arg \max_{\|\delta\| \leq \varepsilon}$ of the action change (for logistic scores, a move of length ε along the weight vector of the gate nearest its threshold), or *Gaussian*, $\delta \sim \mathcal{N}(0, \varepsilon^2 I)$. For a discrete final action a , $\text{AMS}(\varepsilon) = \mathbb{E}_x[\text{TV}(a(x), a(x + \delta))] = P(a(x + \delta) \neq a(x))$, the per-point total variation, equivalently the flip mass. For a continuous actuator signal u , $\text{AMS}(\varepsilon) = \mathbb{E}_x|u(x + \delta) - u(x)|$. The population-level total variation $|\mathbb{E}a(x + \delta) - \mathbb{E}a(x)|$ is *not* used: flips in the two directions cancel and it is near zero for every pipeline. Report $\text{AMS}(\varepsilon)/\varepsilon$ over a grid of ε ; Lemma 2 predicts a constant $\approx \sum_h 2p_h(\tau_h)$ for the crisp pipeline and a constant $\leq L_D H L$ for the fuzzy signal.

7.2 Generator

The two-regime stream of Appendix D with $a = 0.01$, $b = 0.04$ ($\pi = (0.8, 0.2)$, $\lambda = 0.95$), emission means 0.5 apart with unit covariance so that one observation is weakly informative about the regime, and a logistic label link $P(d = 1 | o, s) = \sigma(3 o_{k(s)})$ with $k(0) = 1$, $k(1) = 2$ so that the per-regime heads are well specified. BSF-S1 is trained with regime labels (the oracle setting of Section 8); both models are temperature-scaled on a held-out split.

7.3 E1: Regime-shift stress (Lemma 1(i))

Evaluation	Memoryless ECE	BSF-S1 ECE
Stationary stream, same π	0.025	0.009
Time steps inside regime 0	0.065	0.015
Time steps inside regime 1 (the rare regime)	0.156	0.017
Shifted stream $\pi' = (0.2, 0.8)$, stale A	0.109	0.010

Table 2: E1. The memoryless head passes the stationary audit and is miscalibrated during every sojourn in the rare regime and after the shift; BSF-S1 stays calibrated with a transition matrix that is stale on the shifted stream. Accuracy on the shifted stream: memoryless 0.666, BSF-S1 0.807.

7.4 E2: Trajectory calibration (Lemma 1(ii)–(iv))

Hop-level ECE of the memoryless head is 0.021 on the stream and 0.021 on its random shuffle: identical, as Lemma 1(iv) requires. The realised variance of its window error count is 1.877 times the binomial baseline that hop-level calibration implies; equation (6) evaluated at the head’s measured regime error rates $e = (0.177, 0.376)$ predicts 1.941. The empirical lag covariance of errors matches $\lambda^k \text{Var}_\pi(e)$ of (5) for $k = 1, \dots, 20$ with relative L^2 error 0.111.

Model	$\hat{\varphi}$	KS p -value	TCE verdict
Memoryless (implied joint: independence)	1.764	3.1×10^{-36}	fail
BSF-S1 (implied joint: FFBS)	1.119	0.04	pass

Table 3: E2, $w = 50$. Both models have small hop-level ECE; only the memoryless head’s implied joint is rejected.

7.5 E3: Cliff cascade (Lemma 2)

Three logistic gates on five-dimensional Gaussian evidence, $\tau = \frac{1}{2}$, product t-norm, identity or regularised-centroid defuzzifier. Single hop, adversarial perturbation: crisp $\text{FM}(\varepsilon)/\varepsilon$ is 0.740, 0.742, 0.739, 0.777, 0.784 for $\varepsilon \in \{0.002, 0.005, 0.01, 0.02, 0.05\}$ against the predicted limit

$2p(0) = 0.798$; fuzzy flip mass is exactly 0 and $\text{AMS}/\varepsilon = 0.207$ at every ε , below the bound $L = \frac{1}{4}$. Sequential cascade: after a crisp gate flips, the next gate flips with probability 0.484 independent of ε , while the fuzzy hop’s membership moves by 0.0003 at $\varepsilon = 0.01$. Coupling, $H = 3$: the log-log slope of $P(\text{all flip})$ against ε is 1.00 with coincident crossing points, 1.79 with crossing points spread by a Gaussian of width 0.3 (between the two regimes at these ε), and 2.97 under independent perturbations, against the predicted 1, H and H . The full-pipeline check with a Mamdani actuator and both perturbation models is `experiments/verify_lemma2.py`: the crisp pipeline’s flip mass is 6–10 $\times$ the fuzzy pipeline’s at every ε under both noise models, and the population total variation is below 10^{-3} for both, as the definition anticipates.

7.6 E4: Composition algebra

Rule “ A and B ” evaluated with A entered once and twice. Idempotence gap $\mathbb{E}|T(a, a, b) - T(a, b)|$: minimum 0.000, product 0.084; the downstream decision at threshold $\frac{1}{2}$ flips on 6.8% of inputs under product and on none under minimum. Naive-Bayes multiplication of probabilities is numerically the product case; the difference is that it *asserts independence* of two copies of one predicate.

7.7 E5: Type-2 separation

Two borderline populations with mean degree $\frac{1}{2}$: sharp, elicited from $\text{Beta}(50, 50)$, and diffuse, from $\text{Beta}(1, 1)$. The point head emits 0.502 and 0.507: the same object. The type-2 head fits $\text{Beta}(49, 49)$ and $\text{Beta}(1.0, 1.0)$. CRPS on the diffuse population: point head 0.251 (the MAE, $\rightarrow \frac{1}{4}$), type-2 head 0.167 ($\rightarrow \frac{1}{6}$); on the sharp population 0.039 and 0.027. A single number cannot say whether the degree is known; Q can, and is rewarded for it by a proper score.

7.8 Baselines and ablations not yet run

Exponential smoothing of the memoryless score; an HMM with hand-crafted Gaussian emissions; BSF-S1 with $A = I$, with μ thresholded, and with Q collapsed to its mean. These are the natural next step for a replication.

8 Threats to Validity and Open Problems

Jev is closed. All statements are about the class of memoryless typed calibrated heads as publicly described; nothing here is an empirical refutation of Jev. Two public facts limit what we can assert. The Score primitive returns a bounded scalar, so some vagueness can be expressed by schema choice, though as a single number rather than a type-2 pair. The State object passed to each call may include whatever history the integrator chooses to serialise, so a harness can approximate memory by prompt construction. Neither changes the structural claim: no exact recursion over a latent regime exists inside the model, and no membership-plus-uncertainty pair exists in the output type.

Oracle regime labels. In Section 7 the regime head and the initial transition estimate use regime labels that a real deployment would not have. The claim being tested is what the filter does *given* an emission model, not whether regimes are identifiable from unlabelled streams; the latter is the K -identifiability problem of Section 6 and is left open.

Membership ground truth. Elicited degrees depend on the elicitation protocol. The type-1/type-2 renormalisation across regimes (Section 6.3) is an open semantic question, stated as such: the architecture is designed so that the question is deferred to the actuator rather than answered implicitly by the model.

Filter misspecification. A wrong (A, g, h) yields exact inference on a wrong model. The surprise monitor is a specification, not a guarantee.

Honesty anchor. We make no claim of escaping the approximation regime. Learning A, g, h, μ, Q is statistical estimation and inherits every failure mode of the underlying network. What changes is the division of labour: *inference*—the recursion (8), the t-norm composition, the single defuzzification—is exact given the model. The paper’s claim is that the approximation should live in the parameters, not in the algebra.

9 Discussion

What changes for benchmarking. A benchmark that reports ECE alone certifies $\bar{e}$ and nothing about φ or about action stability. We propose that decision-model benchmarks report TCE and AMS alongside ECE, on streams with a planted regime structure and on evidence with a planted common mode. Both metrics can be computed from the outside, against a closed model, using only its returned probabilities and a harness. Two models with equal ECE are not equally safe, and a benchmark that cannot tell them apart is not measuring safety.

Industrial precedent as evidence. Fuzzy control, probabilistic forecasting, regime-switching econometrics and POMDP planning are four traditions with disjoint literatures, disjoint communities and disjoint failure histories. Each converged on carrying the uncollapsed object to the point of action. When independent traditions agree, the agreement is evidence about the problem rather than about the traditions.

Positioning. This is not an anti-neural argument. In BSF-S1 the network does all the perception: it parameterises A, g, h, μ and Q , and none of those objects could be hand-built at the scale of real evidence. What the network should not do is the reasoning that a fifty-year-old algebra already does exactly. The synthesis is neuro-probabilistic-fuzzy: networks parameterise, algebras compose, and the actuator collapses.

Open problems. (1) Type-2 filter semantics: the correct combination rule for regime-indexed memberships. (2) Learned parametric t-norms (Frank, Hamacher families) with interpretability preserved. (3) Belief-aware thresholds, τ as a function of b_t , and their interaction with Lemma 2. (4) A theory of trajectory calibration as a proper-scoring object in its own right.

10 Conclusion

Two primitives, two lemmas, one principle, one architecture. System-one decision models are the right idea—software needs an interface it can depend on, and typed calibrated probabilities are a far better interface than strings—but the type signature is not yet finished. Give these models a memory of the world’s hidden state and a language for its borderline cases, carry both to the point of action, and collapse nothing until that moment.

Acknowledgments and AI-usage statement

The author used Claude Fable 5.1 (Anthropic) as a writing and coding assistant: it expanded outline bullets into prose, drafted the experiment code, and typeset the manuscript under the author’s direction. All claims, lemmas, proofs, experimental design, interpretation of results and errors are the author’s sole responsibility. No AI system is an author of this work. Code, results and the full revision history are at <https://github.com/dnakhoa/jev-deferred-crispification>.

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A Notation

$o_t, d_t \in D, s_t \in \{1..K\}$	evidence, typed decision target, latent regime
$A, \pi, \lambda, \lambda_i$	transition matrix, its stationary distribution, its non-unit eigenvalues
$g(o), h_s(o)$	regime head $\approx P_{\text{tr}}(s \mid o)$; per-regime decision head $\approx P(d \mid o, s)$
Δb_t	scaled likelihood $\propto g(o_t)/\pi$ (composable)
b_t	filtered belief $P(s_t \mid o_{1:t})$ (not composable)
$\mu \in [0, 1]^{K \times M}, Q$	per-regime membership over M predicates; type-2 distribution over the degree
τ, T, S, D	threshold, t-norm, t-conorm, defuzzifier
$E_t, N_T, \bar{e}, e_s, \varphi$	error indicator, window count, mean and regime error rates, inflation factor

B Full Proofs

Proof of Lemma 1(ii). Let $e = (e_1, e_2)^\top$ and $\Pi = \text{diag}(\pi)$. By the conditional independences, $P(E_t = 1, E_{t+k} = 1) = \sum_{s, s'} \pi(s)(A^k)_{ss'} e_s e_{s'} = e^\top \Pi A^k e$. Every 2×2 row-stochastic A has eigenvalues 1 and $\lambda = 1 - a - b$, with right eigenvector $\mathbf{1}$ and left eigenvector $\pi^\top$ for eigenvalue 1, so $A^k = \mathbf{1}\pi^\top + \lambda^k(I - \mathbf{1}\pi^\top)$. Then $e^\top \Pi \mathbf{1}\pi^\top e = (\pi^\top e)^2 = \bar{e}^2$ and $e^\top \Pi(I - \mathbf{1}\pi^\top)e = \sum_s \pi(s)e_s^2 - \bar{e}^2 = \text{Var}_\pi(e)$. Subtracting $\mathbb{E}[E_t]\mathbb{E}[E_{t+k}] = \bar{e}^2$ gives (5). For (6), $\text{Var}(N_T) = \sum_t \text{Var}(E_t) + 2 \sum_{t < t'} \text{Cov}(E_t, E_{t'}) = T\bar{e}(1 - \bar{e}) + 2\text{Var}_\pi(e) \sum_{k=1}^{T-1} (T-k)\lambda^k$. Dividing by $T\bar{e}(1 - \bar{e})$ and letting $T \rightarrow \infty$, $\sum_{k \geq 1} \lambda^k = \lambda/(1 - \lambda)$ for $|\lambda| < 1$ gives (7). Sojourn in regime 1 is geometric with success probability a and mean $1/a$. For K regimes with A diagonalisable, replace the two-term decomposition by $A^k = \sum_i \lambda_i^k P_i$ with $\lambda_1 = 1$, $P_1 = \mathbf{1}\pi^\top$. $\square$

Proof of Lemma 1(iii). Write $c_s = 1 - e_s$, so $P(N_T = 0) = \mathbb{E}[\prod_t c_{s_t}]$. Every 2×2 stochastic matrix with $\lambda \geq 0$ admits the representation $A = (1 - \lambda)\mathbf{1}\pi^\top + \lambda I$: at each step the chain redraws s from π with probability $1 - \lambda$ and keeps it with probability λ . Conditioning on the redraw times partitions $[1, T]$ into independent frozen blocks of lengths $\ell_1, \dots, \ell_J$. Within a block, $\mathbb{E}[\prod c] = \sum_s \pi(s)c_s^{\ell_j} \geq (\sum_s \pi(s)c_s)^{\ell_j} = (1 - \bar{e})^{\ell_j}$ by Jensen, since $x \mapsto x^\ell$ is convex on $[0, 1]$, with strict inequality iff $\ell_j \geq 2$ and c is non-constant on the support of π . Multiplying over blocks and taking expectation over the partition gives $P(N_T = 0) \geq (1 - \bar{e})^T$, strict when some block has length at least two with positive probability ($\lambda > 0$) and $\text{Var}_\pi(c) > 0$. $\square$

Proof of Lemma 1(iv). ECE and averaged proper scores are functions of the empirical measure $T^{-1} \sum_t \delta_{(f(o_t), y_t)}$, which is invariant under permutations of t . $\square$

Proof of Lemma 2(i). For the product t-norm, on $[0, 1]^H$,

$$\left| \prod_h a_h - \prod_h b_h \right| = \left| \sum_h \left(\prod_{j < h} b_j \right) (a_h - b_h) \left(\prod_{j > h} a_j \right) \right| \leq \sum_h |a_h - b_h|.$$

For minimum, with $\min a = a_i \leq \min b = b_j$, $\min b - \min a \leq b_i - a_i \leq \max_h |a_h - b_h|$. Combine with the Lipschitz constants of μ_h and D . $\square$

Proof of Lemma 2(ii). Lower bound: for x with $\tau_h - c\varepsilon \leq f_h(x) < \tau_h$, moving distance ε along $\nabla f_h / \|\nabla f_h\|$ raises f_h by at least $c\varepsilon$ (mean value theorem with $\|\nabla f_h\| \geq c$ near the boundary), crossing τ_h ; so $x \in F_\varepsilon$, and the band has P_X -mass $\int_{\tau_h - c\varepsilon}^{\tau_h} p_h \approx c p_h(\tau_h) \varepsilon$. Upper bound: if $|f_h(x) - \tau_h| > L\varepsilon$ for all h then no ε -ball crosses any threshold; the complement has mass at most $\sum_h 2L p_h(\tau_h) \varepsilon$. $\square$

Proof of Lemma 2(iii),(iv). (iii) is (10) with δ_h fixed. (iv): with a common crossing value z^* , the set of z within ε of z^* has P_Z -measure $\approx p_Z(z^*)\varepsilon$, and monotonicity of each ϕ_h in z forces all H thresholds to be crossed together. If the per-hop crossing values $z_h^*(\xi_h)$ have a continuous joint density, all H must fall in a window of width 2ε (measure $\Theta(\varepsilon^{H-1})$) and z must fall in the intersection (measure $\Theta(\varepsilon)$), giving $\Theta(\varepsilon^H)$; under independent per-hop perturbations the flip events are independent and the product of H masses of order ε is again $\Theta(\varepsilon^H)$. $\square$

C Forward-Filter/Backward-Sample for TCE

Run Algorithm 1 to obtain $b_{1:T}$. Sample $s_T \sim b_T$; for $t = T - 1, \dots, 1$ sample $s_t \propto b_t(s)A_{s, s_{t+1}}$; draw $E_t \sim \text{Bernoulli}(1 - h_{s_t}(o_t)[d_t^*])$; repeat R times to obtain the implied distribution of N_w . The memoryless head has no joint model beyond independence, so its implied distribution is Poisson-binomial with its own p_t .

D Synthetic Generators

```

1: function REGIMESTREAM( $T, a, b, n$ )
2:    $A \leftarrow \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ ;  $\pi \leftarrow (b, a)/(a+b)$ ;  $s_0 \sim \pi$ ;  $s_{t+1} \sim A[s_t, :]$ 
3:    $o_t \sim \mathcal{N}(m_{s_t}, \Sigma_{s_t})$  in  $\mathbb{R}^n$ ;  $d_t \sim \text{Bernoulli}(\sigma(w_{s_t} \cdot o_t + c_{s_t}))$   $\triangleright$  logistic link so heads are
   well specified
4:   return ( $o, d, s$ )
5: end function
6: function BORDERLINEPOPULATION( $N$ , sharp)
7:    $o \sim$  uniform on the decision-boundary band;  $(\alpha, \beta) \leftarrow (50, 50)$  if sharp else  $(1, 1)$ 
8:    $\mu_m^{\text{elicited}} \sim \text{Beta}(\alpha, \beta)$  per predicate  $m$ 
9:   return ( $o, \mu^{\text{elicited}}$ )
10: end function

```

Sweep $\lambda \in \{0, 0.5, 0.9, 0.99\}$ via (a, b) ; choose (w_s, c_s) so that $P(d \mid o, s)$ differs across s , otherwise $\Delta(o) \equiv 0$ and E1 is vacuous by design.